HADOOP + SPARK — QUICK START

Air Quality Big Data Platform | 23AID302 | Docker cluster cheat sheet

A. Start Hadoop (HDFS cluster)

Step 1 — Open Docker Desktop

```
Double-click the Docker Desktop icon.
Wait ~30-60s until the whale icon in the system tray stops animating.
```

Step 2 — Open a terminal

```
PowerShell or Command Prompt
```

Step 3 — Start the 5 containers

```
docker start aq-namenode aq-datanode1 aq-datanode2 aq-datanode3 aq-datanode4
```

You should see all 5 names printed back.

Step 4 — Wait for safe mode OFF

```
docker exec aq-namenode hdfs dfsadmin -safemode get
```

Re-run every few seconds until it says **Safe mode is OFF** (usually < 1 min).

Step 5 — Open the dashboard in your browser

```
http://localhost:9870
```

Shows Live Nodes: 4. Utilities → Browse the file system to see uploaded data.

Step 6 — (Optional) Health check from terminal

```
docker exec aq-namenode hdfs dfsadmin -report
```

Look for **Live datanodes: 4**, 0 corrupt / missing / under-replicated blocks.

B. Start Spark (runs inside the cluster)

Step 7 — Start the Spark client container

```
docker start aq-spark
```

Step 8 — Submit a job

```
docker exec aq-spark /opt/spark/bin/spark-submit /scripts/01_clean_select.py
```

Always use the full path `/opt/spark/bin/spark-submit` — it is not on the container's PATH.

Step 9 — Watch it run

```
http://localhost:4050
```

The Spark UI is only live **while a job is running**. Between jobs the page will look empty or refuse to connect — that is normal, not an error.

C. Stop everything (end of session)

Step 10 — Stop the containers

```
docker stop aq-spark aq-namenode aq-datanode1 aq-datanode2 aq-datanode3 aq-datanode4
```

Data is preserved — stopping containers does not delete anything. Just repeat Section A/B next time to start fresh.

Quick Reference

What	URL / Command
Hadoop NameNode dashboard	`http://localhost:9870`
Spark job UI (only while running)	`http://localhost:4050`
Check HDFS is healthy	`docker exec aq-namenode hdfs dfsadmin -report`
List files in HDFS	`docker exec aq-namenode hdfs dfs -ls -R /user/hadoop`
See what's running	`docker ps`